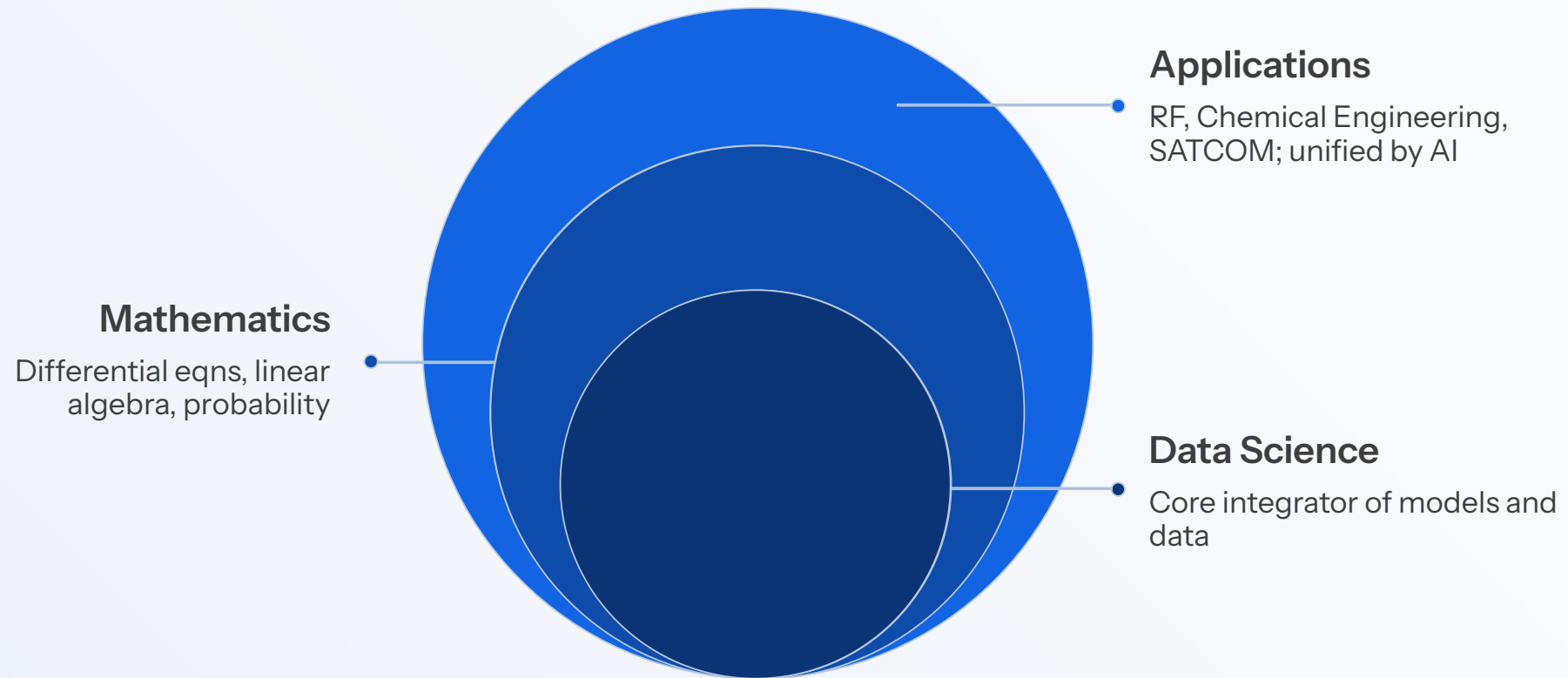


Interdisciplinary Engineering

SATCOM, RF Engineering, Chemical Engineering, and Data Science — built on the same mathematical foundations, converging across aerospace, defense, telecommunications, and AI-driven infrastructure.



A Systems Engineering Perspective



All four disciplines rely on the same core pillars: **mathematical modeling**, signal or process measurement, **optimization**, control systems, and **data-driven decision making**. Their convergence is not coincidental — it is structural.

SATCOM ↔ RF Engineering

This is the most direct relationship. **SATCOM is essentially an application of RF engineering** — every satellite link depends on the physics and hardware that RF engineering defines.

These core equations underpin every satellite communication link:

Free-Space Path Loss

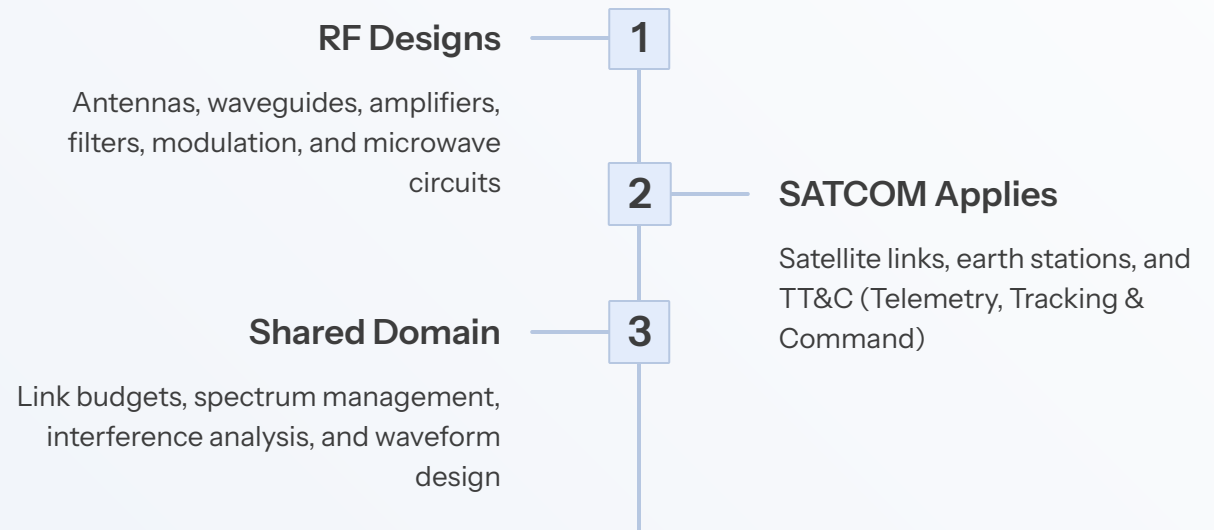
Determines attenuation between a satellite and an earth station

Friis Transmission Equation

Predicts received RF power at the earth station or satellite receiver

Shannon Capacity

Determines the maximum theoretical communication data rate for a link



RF Engineering Chemical Engineering

Although they appear unrelated, these disciplines intersect extensively in **manufacturing, materials science, and thermal management**.



Semiconductor Manufacturing

Chemical engineers handle wafer processing, etching, deposition, CMP, and photoresists. RF engineers design the ICs, MEMS, and microwave devices that result.



Battery Technology

Chemical engineers optimize electrolytes and thermal stability. RF engineers optimize wireless power and battery-powered communication systems.



Thermal Management

Fourier's Law governs heat transfer for satellite payload cooling, RF amplifiers, and data centers — a shared engineering challenge.



Fluid Dynamics

Continuity equations govern satellite cooling loops, fuel systems, and manufacturing processes that underpin RF hardware performance.

Chemical Engineering Data Science

Modern chemical plants generate enormous volumes of sensor data. Data science has become indispensable for making that data actionable.

01

Predictive Maintenance

ML models detect early signs of equipment failure before shutdowns occur

02

Process Optimization

Algorithms find optimal operating conditions to maximize yield and minimize waste

03

Fault Detection

Anomaly detection identifies deviations from normal process behavior in real time

04

Digital Twins

Full virtual replicas of chemical processes enable simulation and what-if analysis

Mass Balance

Machine learning predicts unknown variables within mass balance frameworks, closing gaps in sensor coverage

Reaction Kinetics

Data science estimates reaction constants, reaction order, and optimal operating conditions from experimental data

SATCOM Data Science

Satellites continuously produce telemetry streams — temperature, voltage, current, attitude, RF power, link quality, and GPS position. Data science transforms this raw stream into operational intelligence.

What Satellites Produce

Thermal Data

Temperature readings from payload, battery, and structural components

Power & RF Data

Voltage, current, RF power levels, and link quality metrics

Attitude & Position

GPS position, orbital state vectors, and attitude control data

What Machine Learning Predicts

- **Component failures** before they occur using time-series anomaly detection
- **Orbit anomalies** using LSTM, Prophet, XGBoost, and Random Forest models
- **RF interference** and jamming events through spectral pattern recognition
- **Space weather impacts** on link quality and satellite component lifetimes

RF Engineering ↔ Data Science

Software-defined radios (SDRs) produce continuous streams of I/Q data. Data science makes these streams intelligent and actionable.



Automatic Modulation Classification

ML classifies signal modulation schemes from raw I/Q samples without prior knowledge of the transmitter



RF Fingerprinting

Unique hardware imperfections in transmitters allow ML to identify individual devices from their RF emissions



Spectrum Occupancy Prediction

Forecasting models predict which frequency bands will be occupied, enabling dynamic spectrum access



Jamming Detection via FFT

The Fast Fourier Transform converts I/Q streams to frequency domain, enabling detection in SDR, SATCOM receivers, radar, and electronic warfare systems

SATCOM Chemical Engineering

Satellite systems depend on chemical engineering far more than most engineers realize — from launch to on-orbit operations to end-of-life disposal.



Propulsion

Rocket propellants, reaction engineering, and combustion chemistry directly determine launch capability and orbital maneuvering



Materials

Radiation-resistant polymers, thermal coatings, and composite structures protect satellites in the harsh space environment



Environmental Control

Heat pipes, thermal loops, and phase-change materials maintain satellite electronics within operating temperature bounds



Fuel Cells & Power

Spacecraft power systems, battery chemistry, and hydrogen storage are all electrochemical engineering problems

The Shared Mathematical Foundation

Every discipline in this system draws from the same mathematical toolkit. This table shows where each area of mathematics applies across all four fields.

Mathematical Area	SATCOM	RF Eng.	Chemical Eng.	Data Science
Calculus	✓	✓	✓	✓
Differential Equations	✓	✓	✓	✓
Linear Algebra	✓	✓		✓
Probability & Statistics	✓	✓	✓	✓
Optimization	✓	✓	✓	✓
Fourier Analysis	✓	✓		✓
Control Theory	✓	✓	✓	✓
Graph Theory	✓			

Optimization & AI as the Common Layer

Optimization is central to all four fields. AI now integrates all four disciplines by providing a common computational layer that works across domain boundaries.

Field-Specific Optimization Goals

SATCOM

Minimize link loss and maximize throughput across varying atmospheric conditions

RF Engineering

Maximize antenna gain and signal-to-noise ratio while minimizing power consumption

Chemical Engineering

Maximize reactor yield and minimize energy input across operating conditions

Data Science

Minimize prediction error and maximize model generalizability across datasets

AI Applications by Discipline

SATCOM

Dynamic link optimization and autonomous orbit management

RF Engineering

Automatic signal classification and adaptive beamforming

Chemical Eng.

Predictive process control and digital twin simulation

Data Science

Intelligent decision support and autonomous anomaly response

Intelligent Satellite Mission Digital Twin

A single flagship GitHub project that integrates all four domains — demonstrating **systems thinking** rather than isolated technical skills, directly reflecting how multidisciplinary teams work at NASA, commercial satellite operators, aerospace manufacturers, and defense contractors.



Satellite Engineering

Payload modeling, TT&C simulation, orbital mechanics, and power subsystem design



RF Engineering

Link budget calculations, antenna gain modeling, spectrum analysis, and RF interference simulation



Chemical Engineering

Battery degradation modeling, thermal management, heat transfer simulation, and propellant consumption modeling



Data Science

Predictive maintenance, anomaly detection, telemetry forecasting, and cybersecurity risk scoring